

Sean Lai

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EDUCATION

Santa Clara University | Santa Clara, CA

Sep. 2022 - June 2026

Bachelor of Science in Computer Science and Engineering

Relevant Coursework: Object Oriented Programming, Machine Learning, Compilers, Abstract Data Types and Structures, Algorithms

Yonsei University | Seoul, South Korea

Aug. 2024 - Dec. 2024

Study Abroad Program, Computer Science Focus

PROJECTS

Mimica | [Source code](#) | SCU Technical Excellence Award, Senior Design 2026 Session Award

May. 2026

Voice Cloning Translation Device

- Built a cloud-connected ESP32 consumer device that captures user speech and delivers a voice-cloned translation in a selected language, integrating edge streaming with an **AWS** backend for reliable pipeline processing
- Developed the end-to-end system (**Python**/C++ processing with an **HTML** interface) and containerized the workflow with **Docker** + shell automation to enable reproducible builds, faster iteration, and smoother demos/testing

Homey | [Source code](#)

Dec. 2025

Roommate Matching Application

- Developed a full-stack roommate finder web app with Tinder-style swiping, real-time messaging via Socket.io, and algorithm-based compatibility matching using **React**, **TypeScript**, **Node.js**, **Express**, and **MongoDB**
- Designed group formation system with democratic voting and collaborative profiles using OOP TypeScript; implemented **JWT** authentication, **RESTful APIs**, and **Docker**-containerized database for secure, real-time user management

Netflix Movie Prediction | [Project overview](#) | [Source code](#)

Dec. 2025

Machine Learning Course Final Project

- Built a collaborative-filtering movie recommender trained on the Netflix Prize dataset (100,480,507 ratings; 480,189 users; 17,770 movies) using **Python** (Jupyter), **PyTorch**, **NumPy/Pandas**, **scikit-learn**
- Developed a PyTorch matrix-factorization model trained with Adam on a 3M-rating sample; achieved probe RMSE ≈ 0.784 (~30.7% error reduction); engineered a scalable pipeline to parse raw text logs, persist to Parquet for fast reloads, and batch-generate 2,817,131 qualifying-set predictions

SKILLS / PROFICIENCIES

Languages: Java, C, C++, JavaScript, Python, HTML/CSS, C#, SQL

Frameworks/Libraries: Pandas, NumPy, TensorFlow, PyTorch, MongoDB, React

Tools/Platforms: Unity, Git, Github, Gitlab, AWS, Docker, Kubernetes, Terraform, VS code, Eclipse, Jupyter Notebook

WORK EXPERIENCE

Wireless Intelligent Networks Laboratory | Santa Clara, CA

Sep. 2025 – Current

Research Assistant

- Conduct research on the **QUIC** network protocol, benchmarking throughput, latency, and packet loss performance against traditional protocols such as TCP and UDP using virtual machines and network testbeds
- Investigate how QUIC transport parameters can be adaptively tuned to support multiple concurrent applications sharing a wireless bottleneck
- Combine controlled wireless simulation with **game theoretic** or **learning based optimization** to dynamically adjust QUIC parameters, with the goal of improving latency, jitter, throughput, and fairness across heterogeneous applications

Cequence Cybersecurity | Santa Clara, CA

July 2025 – Sep. 2025

Customer Engineering Intern

- Automated **AWS** infrastructure setup using **Terraform**, **Ansible**, **Docker**, and **CI/CD pipelines**, enabling faster testing and deployment of cybersecurity API gateway plug-ins while supporting scalable multi-user workflows
- Developed infrastructure features for **containerization**, automated resource lifecycle management, and cloud cost optimization, reducing monthly AWS spending for the Customer Engineering team by approximately **10%**
- Collaborated with engineering teams across Australia, India, and Japan to develop, test, and showcase cybersecurity API gateway plug-ins for customer demonstrations.

Santa Clara University Wave+Imaginarium | Santa Clara, CA

Jan. 2024 – Dec. 2024

VR and AR Researcher / Game Developer

- Developed 3D environments, gameplay mechanics, and interactive assets in **Unity** using **C# scripting**, including graphical effects and physics-based object behaviors within a virtual forest environment
- Contributed to the Greenhouse Game Development team by building a 3D first-person PC game set in a live forest virtual space, collaborating on game systems and immersive user experiences
- Partnered with senior engineers and researchers to create a body-positivity virtual experience using **Unity**, **Blender**, and real-time movement tracking technologies